

B.A. / B.Sc. Semester VI Mid-Term Examination 2018–19

MTB 603: Numerical Analysis

Date: 07/03/2019 | **Time:** 1 Hour | **Maximum Marks:** 20

Instructions: Answer Question No. (1) and any three from (2–5). All questions are of equal value (5 Marks each).

Complete Examination Monograph & Verified Solutions

Complete Analytical, Iterative & Theoretical Solutions

Solutions Prepared & Formatted by: **Anushka**

Contains comprehensive line-by-line proofs, error scale examples, order of convergence derivations, Cassini ovals, fixed-point contraction tests, Newton-Raphson iterations, Gauss elimination with partial pivoting, and eigenvalue modulus bounds.

Question 1

[2 + 1 + 2 = 5 Marks]

Part (a): Relative Error vs. Absolute Error

[2 Marks]

Question 1(a)

Show by examples that the relative error is more important than the absolute error.

Mathematical Framework

Let x represent the true (exact) numerical value of a measured or computed physical quantity, and let x_A represent an approximation or empirical observation of x .

1. **Absolute Error (E_a):** Defined as the magnitude of the numerical discrepancy between the true value and its approximation:

$$E_a = |x - x_A| \quad (1)$$

The absolute error E_a carries the same physical dimensions and units as x .

2. **Relative Error (E_r):** Defined as the ratio of the absolute error to the magnitude of the true value x (for $x \neq 0$):

$$E_r = \frac{E_a}{|x|} = \frac{|x - x_A|}{|x|} \quad (2)$$

The relative error E_r is a dimensionless pure number, expressing the fractional discrepancy. It is frequently presented as a percentage error:

$$E_p = E_r \times 100\% \quad (3)$$

Why Absolute Error Alone Fails to Convey Precision

The absolute error E_a fails to account for the **scale** or order of magnitude of the measured quantity. An absolute deviation of 1 cm is utterly imperceptible when measuring the distance between two major cities, but fatal when manufacturing a mechanical wristwatch gear or silicon microprocessor chip. Thus, knowing only E_a without knowing the true magnitude $|x|$ provides zero information regarding whether the measurement is remarkably precise or disastrously erroneous.

Contrasting Real-World Examples

Example 1 (Distance Geodesy vs. Precision Mechanical Component):

- **Measurement A (Long-Distance Highway Survey):**

Let the true distance between two survey stations be $x_A = 10,000$ m (10 km).

Suppose the surveyed approximation is $x_A^* = 10,001$ m.

$$\text{Absolute Error: } E_a = |10,000 - 10,001| = 1 \text{ m} \quad (4)$$

$$\text{Relative Error: } E_r = \frac{1}{10,000} = 10^{-4} = 0.01\% \quad (5)$$

A measurement with only 0.01% error possesses 99.99% accuracy, which is top-tier surveying precision.

- **Measurement B (Machining a Precision Ball Bearing):**

Let the true specified diameter of a miniature ball bearing be $x_B = 2$ mm = 0.002 m.

Suppose the manufactured component measures $x_B^* = 3$ mm = 0.003 m.

$$\text{Absolute Error: } E_a = |0.002 - 0.003| = 0.001 \text{ m} = 1 \text{ mm} \quad (6)$$

$$\text{Relative Error: } E_r = \frac{0.001}{0.002} = 0.5 = 50\% \quad (7)$$

A relative error of 50% represents a complete manufacturing failure.

Critical Analysis: In Measurement B, the absolute error (1 mm = 0.001 m) is **one thousand times smaller** than the absolute error in Measurement A (1 m). If one evaluated quality solely by absolute error, one would falsely conclude that Measurement B is vastly superior to Measurement A. In reality, Measurement A has 99.99% relative accuracy, while Measurement B is defective with 50% error.

Example 2 (Floating-Point Arithmetic & Significant Digits): Consider two computational quantities:

- Let $x = 10^8$ and $x^* = 10^8 + 10$:

$$E_a = 10, \quad E_r = \frac{10}{10^8} = 10^{-7} = 0.00001\% \quad (\approx 7 \text{ correct digits})$$

- Let $y = 10^{-5}$ and $y^* = 2 \times 10^{-5}$:

$$E_a = 10^{-5} = 0.00001, \quad E_r = \frac{10^{-5}}{10^{-5}} = 1.0 = 100\% \quad (0 \text{ correct digits})$$

Although $E_a(y) = 10^{-5}$ is one million times smaller than $E_a(x) = 10$, the calculation for x is highly accurate while the calculation for y has no reliable precision.

Core Takeaway: Supremacy of Relative Error

The relative error $E_r = \frac{E_a}{|x|}$ is fundamentally more important than absolute error because:

1. It is **scale-invariant** and dimensionless, independent of physical units.
2. It directly measures the number of **correct significant digits** in scientific calculations.
3. It allows meaningful comparison of precision across entirely different physical domains and scales.

Part (b): Order of Convergence of Iteration Schemes

[1 Mark]

Question 1(b)

Obtain the order of convergence in two schemes to find a real root of $f(x) = 0$ if errors are related as:

$$(i) \varepsilon_n^2 = 3\varepsilon_{n-1}^3 \quad \text{and} \quad (ii) \varepsilon_{n-1} = \varepsilon_n^{0.5}$$

where ε_n is the error in n^{th} approximation of the root.

Definition of Order of Convergence

Let α denote the true root of $f(x) = 0$, and let $\{x_n\}$ be an iterative sequence converging to α . The error at the n^{th} step is $\varepsilon_n = |x_n - \alpha|$. An iteration method is said to have **order of convergence** $p \geq 1$ if there exists a positive finite constant $C > 0$ (the asymptotic error constant) such that:

$$\lim_{n \rightarrow \infty} \frac{\varepsilon_n}{\varepsilon_{n-1}^p} = C \quad (8)$$

or equivalently, the recurrence relationship satisfies $\varepsilon_n \approx C \varepsilon_{n-1}^p$ for sufficiently large n .

Analysis of Scheme (i): $\varepsilon_n^2 = 3\varepsilon_{n-1}^3$

We are given the error relationship:

$$\varepsilon_n^2 = 3\varepsilon_{n-1}^3 \quad (9)$$

Since errors are non-negative magnitudes ($\varepsilon_n \geq 0$), we take the positive square root on both sides:

$$\varepsilon_n = \left(3\varepsilon_{n-1}^3\right)^{1/2} = \sqrt{3}\varepsilon_{n-1}^{3/2} = \sqrt{3}\varepsilon_{n-1}^{1.5} \quad (10)$$

Dividing both sides by $\varepsilon_{n-1}^{3/2}$:

$$\frac{\varepsilon_n}{\varepsilon_{n-1}^{3/2}} = \sqrt{3} \approx 1.73205 \quad (11)$$

Comparing this directly with the canonical definition $\frac{\varepsilon_n}{\varepsilon_{n-1}^p} = C$:

$$p = \frac{3}{2} = 1.5, \quad C = \sqrt{3}$$

Analysis of Scheme (ii): $\varepsilon_{n-1} = \varepsilon_n^{0.5}$

We are given the recurrence relationship expressing ε_{n-1} in terms of ε_n :

$$\varepsilon_{n-1} = \varepsilon_n^{0.5} = \varepsilon_n^{1/2} \quad (12)$$

To determine the order of convergence, the current error ε_n must be expressed as a function of the prior error ε_{n-1} . Squaring both sides:

$$(\varepsilon_{n-1})^2 = \left(\varepsilon_n^{1/2}\right)^2 \implies \varepsilon_n = 1 \cdot \varepsilon_{n-1}^2 \quad (13)$$

Dividing by ε_{n-1}^2 :

$$\frac{\varepsilon_n}{\varepsilon_{n-1}^2} = 1 \quad (14)$$

Comparing with $\frac{\varepsilon_n}{\varepsilon_{n-1}^p} = C$:

$$p = 2, \quad C = 1$$

Final Results for Order of Convergence

- **Scheme (i):** The order of convergence is $p = \frac{3}{2} = 1.5$ (with asymptotic error constant $C = \sqrt{3}$).
- **Scheme (ii):** The order of convergence is $p = 2$ (**Quadratic Convergence**, with asymptotic error constant $C = 1$).

Part (c): Eigenvalue Inclusion Interval by Brauer's Theorem

[2 Marks]

Question 1(c)

Find the interval which contains the eigen values of a symmetric matrix $A = (a_{ij})_{3 \times 3}$ with:

$$a_{11} = 3, \quad a_{12} = 2, \quad a_{13} = -2, \quad a_{22} = 1, \quad a_{23} = -1, \quad a_{33} = 2$$

using Brauer theorem.

Matrix Formulation

Since A is given to be a symmetric matrix ($A = A^T$), its elements satisfy $a_{ji} = a_{ij}$ for all i, j . Thus:

$$a_{21} = a_{12} = 2, \quad a_{31} = a_{13} = -2, \quad a_{32} = a_{23} = -1$$

The full real symmetric matrix A is:

$$A = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 1 & -1 \\ -2 & -1 & 2 \end{pmatrix} \quad (15)$$

Since A is real symmetric, all of its eigenvalues are guaranteed to be real: $\lambda \in \mathbb{R}$.

Statement of Brauer's Theorem (Ovals of Cassini, 1947)

Let $A = (a_{ij})_{n \times n}$ be an $n \times n$ matrix with complex or real entries. Let R_i denote the deleted row sum of the moduli of the off-diagonal entries in the i^{th} row:

$$R_i = \sum_{\substack{k=1 \\ k \neq i}}^n |a_{ik}| \quad (16)$$

Theorem (Alfred Brauer, 1947): Every eigenvalue λ of A lies in the union of the $\binom{n}{2} = \frac{n(n-1)}{2}$ Cassini ovals:

$$\lambda \in \bigcup_{1 \leq i < j \leq n} K_{ij} \quad (17)$$

where each Cassini region K_{ij} in the complex plane is defined by:

$$K_{ij} = \{z \in \mathbb{C} : |z - a_{ii}| |z - a_{jj}| \leq R_i R_j\} \quad (18)$$

Because A is symmetric, every eigenvalue is real ($\lambda = x \in \mathbb{R}$). Hence, on the real number line, the inclusion region consists of the union of intervals satisfying:

$$|x - a_{ii}| |x - a_{jj}| \leq R_i R_j, \quad 1 \leq i < j \leq 3 \quad (19)$$

Calculation of Row Radii (R_i)

- **Row 1** ($i = 1$): $a_{11} = 3 \implies R_1 = |a_{12}| + |a_{13}| = |2| + |-2| = 2 + 2 = 4$
- **Row 2** ($i = 2$): $a_{22} = 1 \implies R_2 = |a_{21}| + |a_{23}| = |2| + |-1| = 2 + 1 = 3$
- **Row 3** ($i = 3$): $a_{33} = 2 \implies R_3 = |a_{31}| + |a_{32}| = |-2| + |-1| = 2 + 1 = 3$

Derivation of the Three Cassini Intervals

For $n = 3$, there are $\binom{3}{2} = 3$ pairs of indices (i, j) :

1. Pair (1, 2): $a_{11} = 3$, $a_{22} = 1$, and $R_1 R_2 = 4 \times 3 = 12$. The Cassini inequality on the real line is:

$$|x - 3| |x - 1| \leq 12 \quad (20)$$

Analyzing outside the interval $[1, 3]$ (where $(x - 3)(x - 1) \geq 0$):

$$(x - 3)(x - 1) \leq 12 \implies x^2 - 4x + 3 \leq 12 \implies x^2 - 4x - 9 \leq 0$$

The roots of the equation $x^2 - 4x - 9 = 0$ are:

$$x = \frac{4 \pm \sqrt{16 - 4(1)(-9)}}{2} = \frac{4 \pm \sqrt{52}}{2} = 2 \pm \sqrt{13}$$

Since $\sqrt{13} \approx 3.60555$:

$$x_{\min} = 2 - \sqrt{13} \approx -1.6056, \quad x_{\max} = 2 + \sqrt{13} \approx 5.6056$$

For $1 \leq x \leq 3$, $|x - 3| |x - 1| \leq 12$ holds trivially. Hence, the interval is:

$$I_{12} = [2 - \sqrt{13}, 2 + \sqrt{13}] \approx [-1.6056, 5.6056] \quad (21)$$

2. Pair (1, 3): $a_{11} = 3$, $a_{33} = 2$, and $R_1 R_3 = 4 \times 3 = 12$. The Cassini inequality on the real line is:

$$|x - 3| |x - 2| \leq 12 \quad (22)$$

For $x \geq 3$ or $x \leq 2$:

$$(x - 3)(x - 2) \leq 12 \implies x^2 - 5x + 6 \leq 12 \implies x^2 - 5x - 6 \leq 0$$

Factoring the quadratic:

$$(x - 6)(x + 1) \leq 0 \implies -1 \leq x \leq 6$$

Thus, the interval for pair (1, 3) is:

$$I_{13} = [-1, 6] \quad (23)$$

3. Pair (2, 3): $a_{22} = 1$, $a_{33} = 2$, and $R_2 R_3 = 3 \times 3 = 9$. The Cassini inequality on the real line is:

$$|x - 1| |x - 2| \leq 9 \quad (24)$$

For $x \geq 2$ or $x \leq 1$:

$$(x - 1)(x - 2) \leq 9 \implies x^2 - 3x + 2 \leq 9 \implies x^2 - 3x - 7 \leq 0$$

The roots of the equation $x^2 - 3x - 7 = 0$ are:

$$x = \frac{3 \pm \sqrt{9 - 4(1)(-7)}}{2} = \frac{3 \pm \sqrt{37}}{2} \approx \frac{3 \pm 6.0828}{2}$$

Thus, the interval for pair (2, 3) is:

$$I_{23} = \left[\frac{3 - \sqrt{37}}{2}, \frac{3 + \sqrt{37}}{2} \right] \approx [-1.5414, 4.5414] \quad (25)$$

Union of Intervals

By Brauer's Theorem, every eigenvalue λ satisfies $\lambda \in I_{12} \cup I_{13} \cup I_{23}$. The overall bounding interval is determined by the minimum lower bound and maximum upper bound:

$$\text{Lower Bound} = \min \left(2 - \sqrt{13}, -1, \frac{3 - \sqrt{37}}{2} \right) = 2 - \sqrt{13} \approx -1.6056 \quad (26)$$

$$\text{Upper Bound} = \max \left(2 + \sqrt{13}, 6, \frac{3 + \sqrt{37}}{2} \right) = 6 \quad (27)$$

Final Eigenvalue Inclusion Interval by Brauer Theorem

All eigenvalues of the symmetric matrix A lie in the real interval:

$$\lambda \in [2 - \sqrt{13}, 6] \approx [-1.6056, 6.0000] \quad (28)$$

Verification against Exact Eigenvalues

The exact eigenvalues of matrix A , obtained by solving $\det(A - \lambda I) = 0$, are:

$$\lambda_1 \approx -0.2705, \quad \lambda_2 \approx 0.6587, \quad \lambda_3 \approx 5.6119$$

All three eigenvalues lie strictly inside the interval $[-1.6056, 6]$. Moreover, Brauer's theorem yields a substantially tighter interval than Geršgorin's circle theorem, which gives $[-2, 7]$.

Question 2

[5 Marks]

Question 2

Express $x^2 - x - 0.2 = 0$ in the form $x = \phi(x)$ in different ways and pick out the possible forms to be used to solve this equation by fixed point iteration method to find smallest positive real root.

1. Exact Roots of the Given Equation

The given quadratic equation is:

$$x^2 - x - 0.2 = 0 \quad (29)$$

Multiplying by 5 gives the integer equation $5x^2 - 5x - 1 = 0$. By the quadratic formula:

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-0.2)}}{2(1)} = \frac{1 \pm \sqrt{1 + 0.8}}{2} = \frac{1 \pm \sqrt{1.8}}{2} \quad (30)$$

Since $\sqrt{1.8} = \sqrt{\frac{9}{5}} = \frac{3}{\sqrt{5}} \approx 1.3416408$:

$$\alpha_1 = \frac{1 + 1.3416408}{2} \approx \mathbf{1.17082} \quad (\text{Positive Root}) \quad (31)$$

$$\alpha_2 = \frac{1 - 1.3416408}{2} \approx \mathbf{-0.17082} \quad (\text{Negative Root}) \quad (32)$$

The equation has exactly one positive real root: $\alpha = \alpha_1 \approx 1.17082$. Thus, $\alpha \approx 1.17082$ is the unique and smallest positive real root.

2. Fixed Point Iteration Convergence Criterion

To solve $f(x) = 0$ by fixed-point iteration, we rearrange it into:

$$x = \phi(x) \quad (33)$$

The iterative sequence is generated by:

$$x_{n+1} = \phi(x_n), \quad n = 0, 1, 2, \dots \quad (34)$$

Banach Contraction Principle: If $\phi(x)$ is continuously differentiable in an interval I containing the root α , the iteration sequence x_n converges to α for any initial guess $x_0 \in I$ if and only if:

$$|\phi'(\alpha)| < 1 \quad (35)$$

Furthermore, the asymptotic rate of linear convergence is governed by the contraction factor $L = |\phi'(\alpha)|$. If $|\phi'(\alpha)| > 1$, the iteration diverges away from α .

3. Exhaustive Analysis of Different Forms $x = \phi(x)$

Formulation 1 (Direct Linear Isolation): Isolating x from the linear term:

$$x = x^2 - 0.2 \implies \phi_1(x) = x^2 - 0.2$$

Differentiating:

$$\phi_1'(x) = 2x$$

Evaluating at the positive root $\alpha \approx 1.17082$:

$$|\phi_1'(\alpha)| = 2(1.17082) = 2.34164 > 1$$

Status: Diverges. Since $|\phi_1'(\alpha)| > 1$, this formulation cannot be used.

Formulation 2 (Square Root Form): Isolating x^2 and taking the positive square root (since we seek the positive root):

$$x^2 = x + 0.2 \implies x = \sqrt{x + 0.2} = (x + 0.2)^{1/2}$$

Here, $\phi_2(x) = \sqrt{x + 0.2}$. Differentiating:

$$\phi_2'(x) = \frac{1}{2\sqrt{x + 0.2}}$$

At the positive root $\alpha \approx 1.17082$, since $\alpha^2 = \alpha + 0.2$, we have $\sqrt{\alpha + 0.2} = \alpha$:

$$|\phi_2'(\alpha)| = \frac{1}{2\alpha} = \frac{1}{2(1.17082)} \approx \frac{1}{2.34164} \approx \mathbf{0.42705} < 1$$

Status: Converges! The contraction factor is $L_2 \approx 0.427 < 1$.

Formulation 3 (Reciprocal / Division by x): Dividing $x^2 - x - 0.2 = 0$ by x (for $x \neq 0$):

$$x - 1 - \frac{0.2}{x} = 0 \implies x = 1 + \frac{0.2}{x}$$

Here, $\phi_3(x) = 1 + \frac{0.2}{x}$. Differentiating:

$$\phi_3'(x) = -\frac{0.2}{x^2}$$

Evaluating at the positive root $\alpha \approx 1.17082$:

$$|\phi_3'(\alpha)| = \frac{0.2}{\alpha^2} = \frac{0.2}{(1.17082)^2} = \frac{0.2}{1.37082} \approx \mathbf{0.14590} \ll 1$$

Status: Converges Rapidly! The contraction factor is $L_3 \approx 0.146 \ll 1$, ensuring very fast convergence.

Formulation 4 (Factorization of $x(x - 1)$): Rewriting as $x(x - 1) = 0.2$:

$$x = \frac{0.2}{x - 1} \implies \phi_4(x) = \frac{0.2}{x - 1}$$

Differentiating:

$$\phi_4'(x) = -\frac{0.2}{(x - 1)^2}$$

Evaluating at $\alpha \approx 1.17082$ (where $\alpha - 1 \approx 0.17082$):

$$|\phi_4'(\alpha)| = \frac{0.2}{(0.17082)^2} = \frac{0.2}{0.02918} \approx \mathbf{6.854} > 1$$

Status: Diverges. Since $|\phi_4'(\alpha)| \gg 1$, this scheme diverges.

4. Comparative Summary of Formulations

Form	Iteration Function $\phi(x)$	$ \phi'(\alpha) $ at $\alpha \approx 1.1708$	Condition	Behavior
1	$\phi_1(x) = x^2 - 0.2$	2.3416	> 1	Diverges
2	$\phi_2(x) = \sqrt{x + 0.2}$	0.4271	< 1	Converges (Valid)
3	$\phi_3(x) = 1 + \frac{0.2}{x}$	0.1459	$\ll 1$	Converges (Fastest)
4	$\phi_4(x) = \frac{0.2}{x-1}$	6.8540	> 1	Diverges

Picked Forms for Finding the Smallest Positive Real Root

The possible forms that can be used to solve the equation are:

1. **Form 2:** $x_{n+1} = \sqrt{x_n + 0.2}$ (Contractive with $|\phi'| \approx 0.427$)
2. **Form 3:** $x_{n+1} = 1 + \frac{0.2}{x_n}$ (Contractive with $|\phi'| \approx 0.146$, recommended for rapid convergence)

5. Numerical Verification

Starting from initial guess $x_0 = 1.0$:

- **Form 3 Iterations** ($x_{n+1} = 1 + \frac{0.2}{x_n}$):

$$x_1 = 1 + \frac{0.2}{1.0} = 1.200000$$

$$x_2 = 1 + \frac{0.2}{1.2} = 1.166667$$

$$x_3 = 1 + \frac{0.2}{1.166667} = 1.171429$$

$$x_4 = 1 + \frac{0.2}{1.171429} = 1.170732$$

$$x_5 = 1 + \frac{0.2}{1.170732} = \mathbf{1.170833} \approx \alpha$$

Within 5 steps, the iteration achieves 4 decimal digits of precision.

Question 3

[5 Marks]

Question 3

Using Newton-Raphson method, find a real root of the equation $x^3 - 5x - 2 = 0$, correct to three decimal places.

1. Algebraic Analysis & Root Localization

Let the given function be:

$$f(x) = x^3 - 5x - 2 = 0 \quad (36)$$

Evaluating $f(x)$ at consecutive integer points:

$$f(0) = 0^3 - 5(0) - 2 = -2$$

$$f(1) = 1^3 - 5(1) - 2 = -6$$

$$f(2) = 2^3 - 5(2) - 2 = 8 - 10 - 2 = -4 < 0$$

$$f(3) = 3^3 - 5(3) - 2 = 27 - 15 - 2 = +10 > 0$$

Since $f(2) < 0$ and $f(3) > 0$, by the Intermediate Value Theorem, a real root lies in the open interval $(2, 3)$.

Algebraic Factorization: Testing negative integers reveals:

$$f(-2) = (-2)^3 - 5(-2) - 2 = -8 + 10 - 2 = 0$$

Thus, $x = -2$ is an exact rational root. Factoring out $(x + 2)$ by synthetic division:

$$x^3 - 5x - 2 = (x + 2)(x^2 - 2x - 1) = 0$$

The roots of the quadratic factor $x^2 - 2x - 1 = 0$ are:

$$x = \frac{2 \pm \sqrt{4 - 4(1)(-1)}}{2} = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$$

The three exact real roots of the polynomial are:

$$x = -2, \quad x = 1 - \sqrt{2} \approx -0.414214, \quad x = 1 + \sqrt{2} \approx \mathbf{2.414214} \quad (37)$$

We now apply the Newton-Raphson method to locate the root lying in $(2, 3)$, which is $\alpha = 1 + \sqrt{2} \approx 2.414$.

2. Newton-Raphson Iteration Formula

The first derivative of $f(x)$ is:

$$f'(x) = 3x^2 - 5 \quad (38)$$

The Newton-Raphson iteration formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 5x_n - 2}{3x_n^2 - 5} \quad (39)$$

Expressing over a common denominator:

$$x_{n+1} = \frac{x_n(3x_n^2 - 5) - (x_n^3 - 5x_n - 2)}{3x_n^2 - 5} = \frac{2x_n^3 + 2}{3x_n^2 - 5} \quad (40)$$

3. Step-by-Step Iteration Calculations

Since the root lies in $(2, 3)$ and $f(2) = -4$, $f(3) = 10$, we choose the midpoint $x_0 = 2.5$ as our initial approximation:

Iteration 1 ($n = 0$):

$$\begin{aligned} f(x_0) &= f(2.5) = (2.5)^3 - 5(2.5) - 2 = 15.625 - 12.5 - 2 = 1.125000 \\ f'(x_0) &= f'(2.5) = 3(2.5)^2 - 5 = 3(6.25) - 5 = 18.75 - 5 = 13.750000 \\ x_1 &= 2.5 - \frac{1.125000}{13.750000} = 2.5 - 0.081818 = \mathbf{2.418182} \end{aligned}$$

Iteration 2 ($n = 1$): With $x_1 = 2.418182$:

$$\begin{aligned} x_1^2 &= 5.847604, & x_1^3 &= 14.140570 \\ f(x_1) &= 14.140570 - 5(2.418182) - 2 = 14.140570 - 12.090910 - 2 = 0.049660 \\ f'(x_1) &= 3(5.847604) - 5 = 17.542812 - 5 = 12.542812 \\ x_2 &= 2.418182 - \frac{0.049660}{12.542812} = 2.418182 - 0.003959 = \mathbf{2.414223} \end{aligned}$$

Iteration 3 ($n = 2$): With $x_2 = 2.414223$:

$$\begin{aligned} x_2^2 &= 5.828473, & x_2^3 &= 14.071233 \\ f(x_2) &= 14.071233 - 5(2.414223) - 2 = 14.071233 - 12.071115 - 2 = 0.000118 \\ f'(x_2) &= 3(5.828473) - 5 = 17.485419 - 5 = 12.485419 \\ x_3 &= 2.414223 - \frac{0.000118}{12.485419} = 2.414223 - 0.000009 = \mathbf{2.414214} \end{aligned}$$

Iteration 4 ($n = 3$): With $x_3 = 2.414214$:

$$f(x_3) \approx 0.000000, \quad x_4 = \mathbf{2.414214}$$

4. Summary Table of Iterates

Iteration (n)	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	2.500000	1.125000	13.750000	2.418182
1	2.418182	0.049660	12.542812	2.414223
2	2.414223	0.000118	12.485419	2.414214
3	2.414214	0.000000	12.485281	2.414214

The iterates x_2 and x_3 coincide to five decimal places (2.41421).

Final Root Correct to Three Decimal Places

The real root of $x^3 - 5x - 2 = 0$ correct to three decimal places is:

$$\mathbf{x \approx 2.414} \quad (41)$$

(Exact value: $1 + \sqrt{2} \approx 2.41421356 \dots$)

Question 4

[5 Marks]

Question 4

Solve the system of equations:

$$\begin{aligned}x + 2y + z &= 0 \\2x + 2y + 3z &= 3 \\-x - 3y &= 2\end{aligned}$$

by Gauss elimination method with pivoting.

1. Matrix Formulation

The given system of 3 linear equations in 3 unknowns is:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 2 & 3 \\ -1 & -3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} \quad (42)$$

The augmented matrix is:

$$[A^{(0)} \mid \mathbf{b}^{(0)}] = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 2 & 3 & 3 \\ -1 & -3 & 0 & 2 \end{array} \right) \quad (43)$$

2. Step 1: Pivoting in Column 1 and Forward Elimination

We examine the elements in column 1 in rows 1, 2, and 3:

$$\max(|a_{11}|, |a_{21}|, |a_{31}|) = \max(|1|, |2|, |-1|) = 2 = |a_{21}|$$

The largest element in magnitude is $a_{21} = 2$ in **Row 2**. We swap **Row 1 and Row 2** ($R_1 \leftrightarrow R_2$):

$$\left(\begin{array}{ccc|c} \mathbf{2} & 2 & 3 & 3 \\ 1 & 2 & 1 & 0 \\ -1 & -3 & 0 & 2 \end{array} \right) \quad (44)$$

The pivot element is $a_{11} = 2$.**Elimination of x :**

- **Row 2:** Multiplier $m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{2}$.

Operation: $R_2 \leftarrow R_2 - \frac{1}{2}R_1$:

$$\begin{aligned}a_{21}^{(1)} &= 1 - \frac{1}{2}(2) = 0 \\a_{22}^{(1)} &= 2 - \frac{1}{2}(2) = 2 - 1 = 1 \\a_{23}^{(1)} &= 1 - \frac{1}{2}(3) = 1 - \frac{3}{2} = -\frac{1}{2} \\b_2^{(1)} &= 0 - \frac{1}{2}(3) = -\frac{3}{2}\end{aligned}$$

- **Row 3:** Multiplier $m_{31} = \frac{a_{31}}{a_{11}} = \frac{-1}{2} = -\frac{1}{2}$.
Operation: $R_3 \leftarrow R_3 - \left(-\frac{1}{2}\right) R_1 = R_3 + \frac{1}{2} R_1$:

$$\begin{aligned} a_{31}^{(1)} &= -1 + \frac{1}{2}(2) = 0 \\ a_{32}^{(1)} &= -3 + \frac{1}{2}(2) = -3 + 1 = -2 \\ a_{33}^{(1)} &= 0 + \frac{1}{2}(3) = \frac{3}{2} \\ b_3^{(1)} &= 2 + \frac{1}{2}(3) = 2 + \frac{3}{2} = \frac{7}{2} \end{aligned}$$

After Step 1, the augmented matrix becomes:

$$[A^{(1)} \mid \mathbf{b}^{(1)}] = \left(\begin{array}{ccc|c} 2 & 2 & 3 & 3 \\ 0 & 1 & -\frac{1}{2} & -\frac{3}{2} \\ 0 & -2 & \frac{3}{2} & \frac{7}{2} \end{array} \right) \quad (45)$$

3. Step 2: Pivoting in Column 2 and Forward Elimination

We examine the elements in column 2 on and below the main diagonal (rows 2 and 3):

$$\max \left(\left| a_{22}^{(1)} \right|, \left| a_{32}^{(1)} \right| \right) = \max (|1|, |-2|) = 2 = |-2|$$

The largest element in magnitude is in **Row 3**. We swap **Row 2 and Row 3** ($R_2 \leftrightarrow R_3$):

$$\left(\begin{array}{ccc|c} 2 & 2 & 3 & 3 \\ 0 & -2 & \frac{3}{2} & \frac{7}{2} \\ 0 & 1 & -\frac{1}{2} & -\frac{3}{2} \end{array} \right) \quad (46)$$

The pivot element is $a_{22} = -2$.

Elimination of y : The multiplier for Row 3 is:

$$m_{32} = \frac{a_{32}}{a_{22}} = \frac{1}{-2} = -\frac{1}{2}$$

Operation: $R_3 \leftarrow R_3 - \left(-\frac{1}{2}\right) R_2 = R_3 + \frac{1}{2} R_2$:

$$\begin{aligned} a_{32}^{(2)} &= 1 + \frac{1}{2}(-2) = 1 - 1 = 0 \\ a_{33}^{(2)} &= -\frac{1}{2} + \frac{1}{2} \left(\frac{3}{2} \right) = -\frac{1}{2} + \frac{3}{4} = \frac{1}{4} \\ b_3^{(2)} &= -\frac{3}{2} + \frac{1}{2} \left(\frac{7}{2} \right) = -\frac{3}{2} + \frac{7}{4} = \frac{1}{4} \end{aligned}$$

The system is now reduced to upper triangular form:

$$[U \mid \mathbf{c}] = \left(\begin{array}{ccc|c} 2 & 2 & 3 & 3 \\ 0 & -2 & \frac{3}{2} & \frac{7}{2} \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} \end{array} \right) \quad (47)$$

4. Back-Substitution

We solve the triangular system sequentially:

1. **From Row 3:**

$$\frac{1}{4}z = \frac{1}{4} \implies \mathbf{z} = \mathbf{1}$$

2. **From Row 2:**

$$-2y + \frac{3}{2}z = \frac{7}{2}$$

Substituting $z = 1$:

$$-2y + \frac{3}{2}(1) = \frac{7}{2} \implies -2y = \frac{7}{2} - \frac{3}{2} = \frac{4}{2} = 2 \implies \mathbf{y} = -1$$

3. **From Row 1:**

$$2x + 2y + 3z = 3$$

Substituting $y = -1$ and $z = 1$:

$$2x + 2(-1) + 3(1) = 3 \implies 2x - 2 + 3 = 3 \implies 2x + 1 = 3 \implies 2x = 2 \implies \mathbf{x} = \mathbf{1}$$

5. Verification in Original Equations

$$\text{Equation 1: } x + 2y + z = 1 + 2(-1) + 1 = 1 - 2 + 1 = 0 \quad \checkmark$$

$$\text{Equation 2: } 2x + 2y + 3z = 2(1) + 2(-1) + 3(1) = 2 - 2 + 3 = 3 \quad \checkmark$$

$$\text{Equation 3: } -x - 3y = -(1) - 3(-1) = -1 + 3 = 2 \quad \checkmark$$

Exact Solution Vector

The unique solution to the linear system is:

$$\mathbf{x} = \mathbf{1}, \quad \mathbf{y} = -1, \quad \mathbf{z} = 1 \quad (48)$$

or in vector form: $(x, y, z)^T = (1, -1, 1)^T$.

Question 5

[5 Marks]

Question 5

Prove that the largest eigen value in modulus of a square matrix can't exceed the largest sum of the moduli of elements along any row or column.

1. Formal Theorem Statement

Let $A = (a_{ij})_{n \times n}$ be an arbitrary square matrix of order n with entries in $\mathbb{C}$ (or $\mathbb{R}$). Let λ denote any eigenvalue of A . Define:

$$R = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| \quad (\text{Maximum absolute row sum, denoted as } \|A\|_\infty) \quad (49)$$

$$C = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| \quad (\text{Maximum absolute column sum, denoted as } \|A\|_1) \quad (50)$$

Theorem: The modulus of every eigenvalue λ of A satisfies:

$$|\lambda| \leq R \quad \text{and} \quad |\lambda| \leq C \quad (51)$$

Consequently, if $\lambda_{\max}$ is the eigenvalue of maximum modulus, its magnitude (the spectral radius $\rho(A) = \max_k |\lambda_k|$) satisfies:

$$|\lambda_{\max}| \leq \min(R, C) \leq R, C \quad (52)$$

That is, the largest eigenvalue in modulus cannot exceed the largest sum of the moduli of elements along any row or column.

2. Proof for the Row Sum Bound ($|\lambda| \leq R$)

Proof. Let λ be an arbitrary eigenvalue of A . By definition of an eigenvalue, there exists a non-zero column eigenvector $\mathbf{x} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \neq \mathbf{0} \quad (53)$$

such that:

$$A\mathbf{x} = \lambda\mathbf{x} \quad (54)$$

Writing this vector equation in component form for each row $i \in \{1, 2, \dots, n\}$:

$$\sum_{j=1}^n a_{ij}x_j = \lambda x_i, \quad i = 1, 2, \dots, n \quad (55)$$

Since $\mathbf{x} \neq \mathbf{0}$, at least one component of $\mathbf{x}$ is non-zero. Let x_k be a coordinate of $\mathbf{x}$ possessing the maximum absolute value (i.e., the infinity norm of $\mathbf{x}$):

$$|x_k| = \max_{1 \leq j \leq n} |x_j| = \|\mathbf{x}\|_\infty > 0 \quad (56)$$

Now consider the k^{th} scalar equation of system (25):

$$\lambda x_k = \sum_{j=1}^n a_{kj}x_j \quad (57)$$

Taking the absolute value (modulus) of both sides:

$$|\lambda x_k| = |\lambda| |x_k| = \left| \sum_{j=1}^n a_{kj} x_j \right| \quad (58)$$

Applying the generalized triangle inequality for complex numbers ($|\sum z_i| \leq \sum |z_i|$):

$$\left| \sum_{j=1}^n a_{kj} x_j \right| \leq \sum_{j=1}^n |a_{kj} x_j| = \sum_{j=1}^n |a_{kj}| |x_j| \quad (59)$$

By the definition of x_k , we have $|x_j| \leq |x_k|$ for every $j = 1, 2, \dots, n$. Substituting this upper bound into the right-hand sum:

$$\sum_{j=1}^n |a_{kj}| |x_j| \leq \sum_{j=1}^n |a_{kj}| |x_k| = |x_k| \sum_{j=1}^n |a_{kj}| \quad (60)$$

Combining inequalities (28), (29), and (30):

$$|\lambda| |x_k| \leq |x_k| \sum_{j=1}^n |a_{kj}| \quad (61)$$

Since $|x_k| > 0$, we divide both sides by $|x_k|$:

$$|\lambda| \leq \sum_{j=1}^n |a_{kj}| \quad (62)$$

The sum on the right-hand side is the sum of the moduli of elements in the k^{th} row of A . This particular row sum is bounded above by the maximum row sum taken across all rows $i \in \{1, 2, \dots, n\}$:

$$\sum_{j=1}^n |a_{kj}| \leq \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| = R \quad (63)$$

Therefore:

$$|\lambda| \leq R = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| \quad (64)$$

Since this holds for every eigenvalue λ , it holds in particular for the eigenvalue of maximum modulus $\lambda_{\max}$. $\square$

3. Proof for the Column Sum Bound ($|\lambda| \leq C$)

We establish the column bound using two equivalent, standard approaches:

Method A (Via Matrix Transpose A^T):

Proof. Recall the fundamental theorem of determinants: for any square matrix A and its transpose A^T , $\det(A^T) = \det(A)$. Therefore:

$$\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I) \quad (65)$$

Hence, A and A^T have the identical characteristic polynomial and identical eigenvalues λ .

Applying the row sum result proven in Section 2 to the transpose matrix A^T : every eigenvalue λ of A^T (and thus of A) satisfies:

$$|\lambda| \leq \max_{1 \leq i \leq n} \sum_{j=1}^n |(A^T)_{ij}| \quad (66)$$

By the definition of the matrix transpose, $(A^T)_{ij} = a_{ji}$. Thus, the i^{th} row sum of A^T is the i^{th} column sum of A :

$$\sum_{j=1}^n |(A^T)_{ij}| = \sum_{j=1}^n |a_{ji}| = \sum_{k=1}^n |a_{ki}| \quad (67)$$

Taking the maximum over all indices $i \in \{1, 2, \dots, n\}$:

$$\max_{1 \leq i \leq n} \sum_{j=1}^n |(A^T)_{ij}| = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| = C \quad (68)$$

Substituting (38) into (36):

$$|\lambda| \leq C = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| \quad (69)$$

□

Method B (Direct Proof via Left Eigenvector):

Proof. Alternatively, let $\mathbf{y}^H \neq \mathbf{0}^H$ be a non-zero row vector satisfying $\mathbf{y}^H A = \lambda \mathbf{y}^H$ (a left eigenvector of A). In components:

$$\sum_{i=1}^n \bar{y}_i a_{ij} = \lambda \bar{y}_j, \quad j = 1, 2, \dots, n$$

Let m be an index such that $|y_m| = \max_{1 \leq j \leq n} |y_j| > 0$. Taking the m^{th} column equation:

$$|\lambda| |y_m| = \left| \sum_{i=1}^n \bar{y}_i a_{im} \right| \leq \sum_{i=1}^n |\bar{y}_i| |a_{im}| \leq |y_m| \sum_{i=1}^n |a_{im}|$$

Dividing by $|y_m| > 0$:

$$|\lambda| \leq \sum_{i=1}^n |a_{im}| \leq \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| = C$$

□

4. Conclusion

Combining the results of Section 2 and Section 3, for any eigenvalue λ :

$$|\lambda| \leq \min \left(\max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|, \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| \right) \quad (70)$$

Hence, the largest eigenvalue in modulus cannot exceed the largest sum of the moduli of elements along any row or column.

Summary of the Spectral Radius Bound

For any square matrix A :

$$\rho(\mathbf{A}) = \max_{\lambda \in \sigma(\mathbf{A})} |\lambda| \leq \min(\|\mathbf{A}\|_\infty, \|\mathbf{A}\|_1) \quad (71)$$

where $\|\mathbf{A}\|_\infty$ is the maximum absolute row sum and $\|\mathbf{A}\|_1$ is the maximum absolute column sum.

Master Summary Table: Examination Solutions (2018–19)

Q.#	Topic	Governing Mathematical Principle / Iteration	Final Result
1(a)	Relative vs. Absolute Error	Scale invariance: $E_r = \frac{ x-x_A }{ x }$ vs. $E_a = x - x_A $	Proved by 2 detailed contrasting examples
1(b)	Order of Convergence	(i) $\varepsilon_n = \sqrt{3} \varepsilon_{n-1}^{1.5} \implies p = 1.5$ (ii) $\varepsilon_n = \varepsilon_{n-1}^2 \implies p = 2$	(i) $p = 1.5$ (ii) $p = 2$
1(c)	Brauer Theorem Eigenvalues	Cassini ovals $ x - a_{ii} x - a_{jj} \leq R_i R_j$ on symmetric matrix A	$\lambda \in [-1.6056, 6.0000]$
2	Fixed-Point Iterations for $x^2 - x - 0.2 = 0$	Contraction criterion $ \phi'(\alpha) < 1$ at $\alpha \approx 1.17082$: $\phi_2(x) = \sqrt{x + 0.2}$ ($L \approx 0.427$) $\phi_3(x) = 1 + \frac{0.2}{x}$ ($L \approx 0.146$)	Forms 2 & 3 converge
3	Newton-Raphson Method for $x^3 - 5x - 2 = 0$	$x_{n+1} = \frac{2x_n^3 + 2}{3x_n^2 - 5}$, starting from $x_0 = 2.5$	$x \approx 2.414$
4	Gauss Elimination with Partial Pivoting	Pivot col 1: swap $R_1 \leftrightarrow R_2$ Pivot col 2: swap $R_2 \leftrightarrow R_3$ Back-substitution	$x = 1, y = -1, z = 1$
5	Eigenvalue Modulus Bound Theorem	$ \lambda \leq \ A\ _\infty = \max_i \sum_j a_{ij} $ $ \lambda \leq \ A\ _1 = \max_j \sum_i a_{ij} $	Proved completely via eigenvectors & A^T

— End of MTB 603 (2018–19) Solutions Monograph —